FACTORING A MAP OF COMPLEXES THROUGH A TERMWISE SPLIT SURJECTION WITH A HOMOTOPY RETRACTION

THE STACKS PROJECT AUTHORS

Editorial note. Counted span is the author's own proof environment, source lines 2970-3033, following the statement at 2944-2968.

Editorial note. No omissions: all four matrices and the homotopy map h are displayed; the final identity is a substitution check.

Allowed standard prerequisites (editorial).

- Dual construction. lemma-make-injective (same chapter, referenced at the first line ~2970). The proof is dual to the injective case; the matrices below make it explicit.

Source reference. Original definition 014G: sources/stacks-derived/pinned-original-source.tex, source lines 2797–2797. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original lemma 013N: sources/stacks-derived/pinned-original-source.tex, source lines 2840–2840. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Lemma 0.1. *Let $\mathcal{A}$ be an additive category. Let $\alpha : K^\bullet \to L^\bullet$ be a morphism of complexes of $\mathcal{A}$. There exists a factorization*

$$K^\bullet \xrightarrow{i} \tilde{K}^\bullet \xrightarrow{\tilde{\alpha}} L^\bullet, \quad K^\bullet \xrightarrow{\alpha} L^\bullet$$

such that

(1) *$\tilde{\alpha}$ is a termwise split surjection (see Definition 014G),*
(2) *there is a map of complexes $s : \tilde{K}^\bullet \to K^\bullet$ such that $s \circ i = id_{K^\bullet}$ and such that $i \circ s$ is homotopic to $id_{\tilde{K}^\bullet}$.*

Moreover, if both $K^\bullet$ and $L^\bullet$ are in $K^+(\mathcal{A})$, $K^-(\mathcal{A})$, or $K^b(\mathcal{A})$, then so is $\tilde{K}^\bullet$.

Proof. Dual to Lemma 013N. Take

$$\tilde{K}^n = K^n \oplus L^{n-1} \oplus L^n$$

Date: stacks-project:derived.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

and we define

$$d^n_{\tilde{K}} = \begin{pmatrix} d^n_K & 0 & 0 \\ 0 & -d^{n-1}_L & \mathrm{id}_{L^n} \\ 0 & 0 & d^n_L \end{pmatrix}$$

in other words $\tilde{K}^\bullet = K^\bullet \oplus C(1_{L^\bullet[-1]})$. Moreover, we set

$$\tilde{\alpha} = \begin{pmatrix} \alpha & 0 & \mathrm{id}_{L^n} \end{pmatrix}$$

which is clearly a split surjection. It is also clear that it defines a morphism of complexes. We define

$$i = \begin{pmatrix} \mathrm{id}_{K^n} \\ 0 \\ 0 \end{pmatrix}$$

so that clearly $\tilde{\alpha} \circ i = \alpha$. We set

$$s = \begin{pmatrix} \mathrm{id}_{K^n} & 0 & 0 \end{pmatrix}$$

so that $s \circ i = \mathrm{id}_{K^\bullet}$. Finally, let $h^n : \tilde{K}^n \to \tilde{K}^{n-1}$ be the map which maps the summand L^{n-1} of $\tilde{K}^n$ via the identity morphism to the summand L^{n-1} of $\tilde{K}^{n-1}$. Then it is a trivial matter to prove that

$$\mathrm{id}_{\tilde{K}^\bullet} - i \circ s = d \circ h + h \circ d$$

which finishes the proof of the lemma. $\square$